

Q: Solve the differential equation by applying $f(p, q) = 0$:

$$(x+y)(p+q)^2 + (x-y)(p-q)^2 = 1$$

Ans:

Given,

$$(x+y)(p+q)^2 + (x-y)(p-q)^2 = 1 \dots (i)$$

$$\text{let, } (x+y) = U^2, \quad (x-y) = V^2$$

we know,

$$p = \frac{z}{x} = \frac{\partial z}{\partial U} \cdot \frac{\partial U}{\partial x} + \frac{\partial z}{\partial V} \cdot \frac{\partial V}{\partial x}$$

$$q = \frac{z}{y} = \frac{\partial z}{\partial U} \cdot \frac{\partial U}{\partial y} + \frac{\partial z}{\partial V} \cdot \frac{\partial V}{\partial y}$$

differentiating w.r.t. 'x' and 'y' we get,

$$(x+y)U^2$$

$$\text{Or, } 1 = 2U \cdot \frac{\partial U}{\partial x}$$

$$\text{Or, } \frac{\partial U}{\partial x} = \frac{1}{2U}$$

$$(x+y) = U^2$$

$$\text{Or, } 1 = 2U \cdot \frac{\partial U}{\partial y}$$

$$\text{Or, } \frac{\partial U}{\partial y} = \frac{1}{2U}$$

$$(x-y) = V^2$$

$$\text{Or, } 1 = 2V \cdot \frac{\partial V}{\partial x}$$

$$\text{Or, } \frac{\partial V}{\partial x} = \frac{1}{2V}$$

$$(x-y)^2 = V^2$$

$$\text{Or, } -1 = 2V \cdot \frac{\partial V}{\partial y}$$

$$\text{Or, } \frac{\partial V}{\partial y} = -\frac{1}{2V}$$

$$\text{So, } p = \frac{1}{2U} \cdot \frac{\partial z}{\partial U} + \frac{1}{2V} \cdot \frac{\partial z}{\partial V} \dots (ii)$$

$$q = \frac{1}{2U} \cdot \frac{\partial z}{\partial U} - \frac{1}{2V} \cdot \frac{\partial z}{\partial V} \dots (iii)$$

Adding (i) & (ii) we get,

$$p+q = \frac{1}{U} \cdot \frac{\partial z}{\partial U} \dots (iv)$$

Subtracting (iii) from (i) we get,

$$p-q = \frac{1}{V} \cdot \frac{\partial z}{\partial V} \dots (v)$$

Using (iv) & (v) in equation (i) we get,

$$U^2 \cdot \left(\frac{1}{U} \cdot \frac{\partial Z}{\partial U} \right)^2 + V^2 \cdot \left(\frac{1}{V} \cdot \frac{\partial Z}{\partial V} \right)^2 = 1$$

$$\left(\frac{\partial Z}{\partial U} \right)^2 + \left(\frac{\partial Z}{\partial V} \right)^2 = 1$$

So, equation becomes

$$P^2 + Q^2 = 1$$

$$\text{Or, } a^2 + b^2 = 1 \quad \therefore b = \sqrt{1-a^2}$$

Let, the solution be:

$$Z = aU + bV + C$$

$$= a\sqrt{x+y} + \sqrt{1-a^2} \sqrt{x-y} + C \quad [\because x+y=U^2, x-y=V^2]$$

So, the solution is: $Z = a\sqrt{x+y} + \sqrt{1-a^2} \sqrt{x-y} + C$ (Ans:)

Q: solve the differential equation by applying $f(z, p, q) = 0$:

$$z^2(p^2x^2 + q^2y^2) = 1$$

Ans:

Given, $z^2(p^2x^2 + q^2y^2) = 1$

$$\text{or, } z^2\left(x^2\left(\frac{\partial z}{\partial x}\right)^2 + y^2\left(\frac{\partial z}{\partial y}\right)^2\right) = 1 \quad \left[\because p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}\right]$$

$$\text{or, } z^2\left(\frac{\partial z}{\partial x/x}\right)^2 + z^2\left(\frac{\partial z}{\partial y/y}\right)^2 = 1 \quad \dots (i)$$

$$\text{kt, } \frac{\partial z}{\partial x} = X \quad \frac{\partial z}{\partial y} = Y$$

$$\text{or, } \ln x = X \quad \text{or, } \ln y = Y$$

$$\text{kt, } u = X + aY$$

$$\text{or, } p = \frac{\partial z}{\partial u} = \frac{\partial z}{\partial X}, q = a \cdot \frac{\partial z}{\partial u} = a \cdot \frac{\partial z}{\partial Y}$$

Substituting the value of 'p' and 'q' in equation we get,

$$z^2\left(\frac{\partial z}{\partial X}\right)^2 + z^2\left(\frac{\partial z}{\partial Y}\right)^2 = 1$$

$$\text{or, } z^2\left(\frac{\partial z}{\partial u}\right)^2 + z^2 \cdot a^2\left(\frac{\partial z}{\partial u}\right)^2 = 1$$

$$\text{or, } z^2\left(\frac{\partial z}{\partial u}\right)^2(1+a^2) = 1$$

$$\text{or, } z \cdot \frac{\partial z}{\partial u} \cdot \sqrt{1+a^2} = 1 \quad [\text{Doing square root in both sides}]$$

$$\text{or, } \int z dz \cdot \sqrt{1+a^2} = \int du$$

$$\text{or, } \frac{z^2}{2} \sqrt{1+a^2} = u + C \quad [\text{Integrating}]$$

$$\text{or, } \frac{z^2}{2} \sqrt{1+a^2} = \ln x + a \ln y + C \quad [\because u = \ln x + a \ln y]$$

$$\therefore \text{ solution is : } \frac{z^2}{2} \sqrt{1+a^2} = \ln x + a \ln y + C \quad (\text{Ans:})$$

Q: Solve the differential equation: $x^2p + y^2q = (x+y)z$

Ans: Given, $x^2p + y^2q = (x+y)z \dots (i)$

Comparing equation (i) with $P + Qq = R$ we get,

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{(x+y)z} \dots (ii)$$

$$\text{Or, } \frac{dx}{x^2} = \frac{dy}{y^2}$$

$$\text{Or, } -\frac{1}{x} = -\frac{1}{y} + C_1 \quad [\text{Integrating}]$$

$$\text{Or, } \frac{1}{x} - \frac{1}{y} = C_1 \dots (iii)$$

$$\text{Again, } \frac{dx/x}{x} = \frac{dy/y}{y} = \frac{dz/z}{(x+y)} = \frac{\frac{dx}{x} + \frac{dy}{y} - \frac{dz}{z}}{(x+y) - (x+y)} \dots (iv)$$

$$\text{So, } \frac{dx}{x} + \frac{dy}{y} - \frac{dz}{z} = 0$$

$$\text{Or, } \ln x + \ln y - \ln z = \ln C_2 \quad [\text{Integrating}]$$

$$\text{Or, } \ln \frac{xy}{z} = \ln C_2$$

$$\text{Or, } \frac{xy}{z} = C_2 \dots (v)$$

Using (iii) & (v) the solution becomes:

$$f\left(\frac{1}{x} - \frac{1}{y}, \frac{xy}{z}\right) = 0 \quad (\text{Ans})$$

Q: solve the differential equation by applying $f(p, x) = g(q, y)$:

$$z^2(p^2 + q^2) = x^2 + y^2$$

Ans:

Given,

$$z^2(p^2 + q^2) = x^2 + y^2$$

$$\text{Or, } z^2 \left\{ \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 \right\} = x^2 + y^2$$

$$\text{Or, } \left(\frac{z \partial z}{\partial x} \right)^2 + \left(\frac{z \partial z}{\partial y} \right)^2 = x^2 + y^2 \dots (i)$$

$$\text{kt, } z \partial z = \partial Z$$

$$\text{Or, } \frac{z^2}{2} = Z$$

$$\text{Now, } P = \frac{\partial Z}{\partial x} = \frac{\partial Z}{\partial z} \cdot \frac{\partial z}{\partial x} = z \cdot \frac{\partial z}{\partial x}$$

$$Q = \frac{\partial Z}{\partial y} = \frac{\partial Z}{\partial z} \cdot \frac{\partial z}{\partial y} = z \cdot \frac{\partial z}{\partial y}$$

So, equation (i) becomes

$$P^2 + Q^2 = x^2 + y^2$$

$$\text{Or, } P^2 - x^2 = y^2 - Q^2$$

$$\text{kt, } P^2 - x^2 = y^2 - Q^2 = a$$

$$\therefore P = \sqrt{x^2 + a}, \quad Q = \sqrt{y^2 - a}$$

Now, the solution will be:

$$dZ = P dx + Q dy$$

$$\text{or, } dZ = \sqrt{x^2 + a} dx + \sqrt{y^2 - a} dy$$

$$\text{or, } Z = \frac{1}{2} x \sqrt{x^2 + a} + \frac{a}{2} \log(x + \sqrt{x^2 + a}) + \frac{1}{2} y \sqrt{y^2 - a} - \frac{a}{2} \log(y + \sqrt{y^2 - a}) + b$$

$$\text{or, } \frac{z^2}{2} = \frac{1}{2}x\sqrt{x^2+a} + \frac{a}{2}\log(x+\sqrt{x^2+a}) + \frac{1}{2}y\sqrt{y^2-a} - \frac{a}{2}\log(y+\sqrt{y^2-a}) + b$$

$$\text{Or, } z = x\sqrt{x^2+a} + y\sqrt{y^2-a} + a\log\frac{x+\sqrt{x^2+a}}{y+\sqrt{y^2-a}} + C \quad [\text{multiplying both sides by '2'}]$$

$$\therefore \text{ solution : } z = x\sqrt{x^2+a} + y\sqrt{y^2-a} + a\log\frac{x+\sqrt{x^2+a}}{y+\sqrt{y^2-a}} + C \quad (\text{Ans:})$$

Q: solve the differential equation: $x(y-z)p - y(x-z)q = z(x-y)$

Ans: Given, $x(y-z)p - y(x-z)q = z(x-y) \dots (i)$

Comparing equation (i) with $Pp + Qq = R$ we get,

$$\frac{dx}{x(y-z)} = -\frac{dy}{y(x-z)} = \frac{dz}{z(x-y)}$$

Taking $l=1, m=1, n=1$

$$\frac{l dx + m dy + n dz}{lP + mQ + nR} = \frac{dx + dy + dz}{xy - zx + yz - xy + xz - yz} = \frac{dx + dy + dz}{0}$$

$$\text{So, } dx + dy + dz = 0$$

$$\text{Or, } x + y + z = C_1 \quad [\text{Integrating}] \dots (ii)$$

Again, $l = \frac{1}{x}, m = \frac{1}{y}, n = \frac{1}{z}$

$$\frac{l dx + m dy + n dz}{lP + mQ + nR} = \frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{y-z - x + z + x - y} = \frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{0}$$

$$\text{So, } \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

$$\text{or, } \ln x + \ln y + \ln z = \ln C_2 \quad [\text{Integrating}]$$

Or, $xyz = c_2 \dots (iii)$

So, the solution be: $f(x+y+z, xyz) = 0$ (Ans:)

Q: Solve: $2z + p^2 + qy + 2y^2 = 0$

Ans: Given, $2z + p^2 + qy + 2y^2 = 0 \dots (i)$

$$\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = q + 4y, \frac{\partial f}{\partial z} = 2, \frac{\partial f}{\partial p} = 2p, \frac{\partial f}{\partial q} = y$$

We know, charpit's method:

$$\frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}} = \frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}}$$

So, $\frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}}$

Or, $\frac{dx}{-2p} = \frac{dp}{2p}$

Or, $-dx = dp$

Or, $dx + dp = 0$

Or, $x + p = a$ [Integrating]

Or, $p = a - x$

Putting the value of p in equation (i) we get,

$$2z + (a-x)^2 + qy + 2y^2 = 0$$

Or, $qy = -\{2z + (a-x)^2 + 2y^2\}$

Or, $q = -\frac{2z + (a-x)^2 + 2y^2}{y}$

So, the solution be:

$$dz = p dx + q dy$$

$$\text{or, } dz = (a-x) dx + \frac{-(2z + (a-x)^2 + 2y^2)}{y} dy$$

$$\text{or, } dz - (a-x) dx = - \frac{2z + (a-x)^2 + y^2}{y} dy - y dy$$

$$\text{or, } dz - (a-x) dx + y dy = - \frac{2z + (a-x)^2 + y^2}{y} dy$$

$$\text{or, } \frac{2 [dz - (a-x) dx + y dy]}{2z + (a-x)^2 + y^2} = - \frac{2}{y} dy$$

In the L.H.S. the numerator is the differential co-efficient of denominator, Hence, integrating we get,

$$\ln [2z + (a-x)^2 + y^2] = \ln \frac{1}{y^2} + \ln b$$

$$\text{or, } \ln [2z + (a-x)^2 + y^2] = \ln \left(\frac{b}{y^2} \right)$$

$$\therefore 2z + (a-x)^2 + y^2 = \frac{b}{y^2}$$

$$\therefore \text{ solution: } 2z + (a-x)^2 + y^2 = \frac{b}{y^2} \quad (\text{Ans:})$$

Q: solve the differential equation by applying $f(p, x) = g(q, y) : (p-x)^2 - x^2 = (q-y)^2 - y^2$

Ans:

Given,

$$(p-x)^2 - x^2 = (q-y)^2 - y^2 \dots (i)$$

$$\text{let, } (p-x)^2 - x^2 = a = (q-y)^2 - y^2$$

$$\therefore p = x \pm \sqrt{x^2 + a}, \quad q = y \pm \sqrt{y^2 + a}$$

We know,

$$dz = p dx + q dy$$

$$\text{or, } dz = (x \pm \sqrt{x^2 + a}) dx + (y \pm \sqrt{y^2 + a}) dy$$

$$\text{or, } \int dz = \int (x \pm \sqrt{x^2 + a}) dx + \int (y \pm \sqrt{y^2 + a}) dy + C$$

$$\text{or, } z = \frac{x^2 + y^2}{2} \pm \frac{1}{2} x \sqrt{x^2 + a} \pm \frac{a}{2} \ln |x + \sqrt{x^2 + a}| \pm \frac{1}{2} y \sqrt{y^2 + a} \pm \frac{a}{2} \ln |y + \sqrt{y^2 + a}| + C$$

or

$$\therefore \text{ solution is: } z = \frac{x^2 + y^2}{2} \pm \frac{1}{2} \left\{ x \sqrt{x^2 + a} + a \ln |x + \sqrt{x^2 + a}| + y \sqrt{y^2 + a} + a \ln |y + \sqrt{y^2 + a}| \right\} + C$$

(Ans:)

Q: solve the differential equation: $(y+z)p - (x+z)q = x-y$

Ans: Given, $(y+z)p - (x+z)q = x-y \dots (i)$

Comparing equation (i) with $Pp + Qq = R$ we get,

$$P = (y+z), Q = -(x+z), R = (x-y)$$

$$\frac{dz}{P} = \frac{dy}{Q} = \frac{dx}{R}$$

$$\frac{dx}{y+z} = \frac{dy}{-(x+z)} = \frac{dz}{x-y} = \frac{l dx + m dy + n dz}{lP + mQ + nR}$$

Taking $l=1, m=1, n=1$,

$$\frac{dx+dy+dz}{y+z-x-z+x-y} = \frac{dx+dy+dz}{0}$$

$$dx+dy+dz = 0$$

$$\text{Or, } \int (dx+dy+dz) = C_1 \quad [\text{Integrating}]$$

$$\text{Or, } x+y+z = C_1$$

Again, Taking $l=x, m=y, n=-z$,

$$\frac{x dx + y dy - z dz}{2x+z-x-y-z-x-yz} = \frac{x dx + y dy - z dz}{0}$$

$$x dx + y dy - z dz = 0$$

$$\text{Or, } \int x dx + \int y dy - \int z dz = C_2$$

$$\text{Or, } x^2 + y^2 - z^2 = C_2$$

$$\therefore \text{ solution: } f(x+y+z, x^2+y^2-z^2) = 0 \quad (\text{Ans:})$$

Q: From the differential equation by eliminating arbitrary functions from the given solution:

$$f(xy^2z, 5x+z^5-2xyz^2)$$

Ans: Given, $f(xy^2z, 5x+z^5-2xyz^2)$

let, $u = xy^2z, v = 5x+z^5-2xyz^2$

we know, $p = \frac{zx}{zx}, q = \frac{zx}{zy}$

Now, $\frac{zu}{zx} = y^2z, \frac{zu}{zy} = 2xyz, \frac{zu}{zz} = xy^2$

$$\frac{zv}{zx} = 5 - 2yz^2, \frac{zv}{zy} = -2xz^2, \frac{zv}{zz} = 5z^4 - 4xyz$$

we know, jacobian w.r.t x, y, z is equal to zero i.e.

$$\begin{vmatrix} p & q & -1 \\ \frac{zu}{zx} & \frac{zu}{zy} & \frac{zu}{zz} \\ \frac{zv}{zx} & \frac{zv}{zy} & \frac{zv}{zz} \end{vmatrix} = 0$$

or,

$$\begin{vmatrix} \frac{zx}{zx} & \frac{zx}{zy} & -1 \\ y^2z & 2xyz & xy^2 \\ 5-2yz^2 & -2xz^2 & 5z^4-4xyz \end{vmatrix} = 0$$

After expanding,

$$\frac{z}{x} \begin{vmatrix} 2yz & xy^2 \\ -2xz^2 & 5z^4 - 4xyz \end{vmatrix} - \frac{z}{y} \begin{vmatrix} y^2z & xy^2 \\ 5 - 2yz^2 & 5z^4 - 4xyz \end{vmatrix} + (-1) \begin{vmatrix} y^2z & 2xyz \\ 5 - 2yz^2 & -2xz^2 \end{vmatrix} = 0$$

$$\text{Or, } \frac{z}{x} \{ 2yz(5z^4 - 4xyz) + xy^2 2xz^2 \} - \frac{z}{y} \{ y^2z(5z^4 - 4xyz) - xy^2(5 - 2yz^2) \} - \{ y^2z(-2xz^2) - 2xyz(5 - 2yz^2) \} = 0$$

$$\text{Or, } \frac{z}{x} (10xyz^5 - 6x^2yz^2) - \frac{z}{y} (5y^2z^5 - 5xy^2 - 2xy^3z^2) - (2xy^2z^3 - 10xyz) = 0$$

$$\text{Or, } 2xyz(5z^4 - 3xyz)p + y^2(5x + 2xyz^2 - 5z^5)q - 2xyz(yz^2 - 5) = 0$$

$$\text{Or, } 2xz(5z^4 - 3xyz)p + y(5x + 2xyz^2 - 5z^5)q - 2xz(yz^2 - 5) = 0 \quad [\text{dividing both sides by } y]$$

$$\therefore \text{ solution: } 2xz(5z^4 - 3xyz)p + y(5x + 2xyz^2 - 5z^5)q - 2xz(yz^2 - 5) = 0 \quad (\text{Ans:})$$

* Bessel's equation & function:

Differential equation of the form $x^2 y'' + xy' + (x^2 - n^2)y = 0$ is called the Bessel's differential equation and the particular solutions of this equation are called Bessel's function of order 'n'.

Let the solution be: $y = \sum_{r=0}^{\infty} a_r x^{m+r}$

$$y' = \sum_{r=0}^{\infty} a_r (m+r) x^{m+r-1}$$

$$y'' = \sum_{r=0}^{\infty} a_r (m+r-1) x^{m+r-2} \quad \left\{ \text{differentiating} \right\}$$

from here we will get $m = n, -n$,

for $m = n$,

$$y_n = a_0 x^n \left\{ 1 - \frac{1}{4(n+1)} x^2 + \frac{1}{4^2 \cdot 2! (n+1)(n+2)} x^4 - \dots + \frac{(-1)^r x^{2r}}{2^r r! \cdot 2^r (n+1)(n+2) \dots (n+r)} \right\}$$

for $m = -n$,

$$y_{-n} = a_0 x^{-n} \left\{ 1 - \frac{1}{4(-n+1)} x^2 + \frac{x^4}{4^2 \cdot 2! (-n+1)(-n+2)} + \dots + \frac{(-1)^r x^{2r}}{2^r r! \cdot 2^r (-n+1)(-n+2) \dots (-n+r)} \right\}$$

If $a_0 = \frac{1}{2^n n!}$ then,

$$y_n = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! (n+r+1)} \left(\frac{x}{2} \right)^{n+2r} = J_n(x)$$

$$y_{-n} = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! (-n+r+1)} \left(\frac{x}{2} \right)^{-n+2r} = J_{-n}(x)$$

*Legendre's function of the second kind $Q_n(x)$:

The second independent solution of Legendre's equation is called the Legendre's function of the second kind, denoted by $Q_n(x)$.

In the descending form:

$$Q_n(x) = \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n+1)} \left[x^{-n-1} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-n-3} + \cdots \right]$$

Unlike $P_n(x)$, the series for $Q_n(x)$ is non-terminating